

NEURUPPIN, Germany

WMO: 102700

Lat: 52.9037N

Lon: 12.8071E

Elev: 128

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.3	13.5	3.3	6.6	10.2	8.1	8.4	14.8	24.8	35.5	22.7	33.5	5.0	70	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	83.2	67.4	80.0	65.7	77.1	64.6	69.0	79.8	67.3	77.0	65.6	74.5	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	93.9	73.5	63.5	88.1	71.7	61.9	83.2	70.4	33.2	80.2	31.8	77.1	30.5	74.7	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.6	16.3	DB	5.0	89.8	9.8	3.5	-2.1	92.2	-7.8	94.3	-13.3	96.2	-20.4	98.7
			WB	4.4	73.9	9.6	1.5	-2.5	75.0	-8.1	75.9	-13.5	76.7	-20.5	77.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	32.8	31.0	38.7	46.1	55.4	60.6	64.6	64.1	57.4	49.5	40.3	35.6	(6)
	DBStd	13.50	9.72	8.64	6.37	6.35	6.52	6.05	5.63	5.50	4.71	6.45	5.95	6.69	(7)
	HDD50	2418	533	532	350	151	23	0	0	0	2	87	294	446	(8)
	HDD65	6325	998	952	814	566	304	161	79	85	233	480	742	911	(9)
	CDD50	1731	0	0	1	35	190	317	452	438	224	72	2	0	(10)
	CDD65	162	0	0	0	0	5	28	66	58	5	0	0	0	(11)
	CDH74	1379	0	0	0	1	54	261	581	444	37	1	0	0	(12)
	CDH80	271	0	0	0	0	1	49	122	97	2	0	0	0	(13)
(14) Wind	WSAvg	7.4	9.3	8.1	8.3	7.4	7.0	6.5	6.4	5.7	6.5	7.1	8.2	8.7	(14)
(15) Precipitation	PrecAvg	23.7	2.0	1.6	1.6	1.2	2.1	2.5	2.9	2.4	2.0	1.8	1.7	2.0	(15)
	PrecMax	34.3	4.6	3.8	3.3	2.6	6.3	5.3	9.2	5.4	5.7	4.8	3.5	3.7	(16)
	PrecMin	16.3	0.0	0.3	0.4	0.0	0.5	0.7	1.0	0.9	0.5	0.4	0.1	0.4	(17)
	PrecStd	4.0	1.1	0.7	0.8	0.6	1.3	1.1	1.8	1.1	1.1	1.0	0.8	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	51.6	58.8	70.7	78.3	85.2	87.1	87.8	77.8	70.1	55.4	52.4	(19)
		MCWB	48.2	47.2	50.7	55.5	63.2	67.4	69.8	70.7	65.4	61.6	51.6	48.7	(20)
	2%	DB	47.9	46.7	53.8	65.3	75.3	80.8	83.6	82.1	74.0	64.5	53.0	49.9	(21)
		MCWB	45.2	43.4	46.9	53.6	61.5	64.7	67.7	67.8	63.5	57.8	50.1	47.6	(22)
	5%	DB	46.0	43.6	50.2	61.0	72.0	76.6	80.2	79.0	69.3	61.2	50.6	47.7	(23)
		MCWB	43.3	40.8	45.1	51.4	59.2	63.0	66.0	66.2	60.6	55.7	48.1	45.6	(24)
	10%	DB	43.8	41.3	47.5	57.2	68.4	72.9	77.2	75.4	66.2	58.6	48.5	45.5	(25)
		MCWB	41.4	38.9	43.3	49.2	57.3	61.2	65.1	64.4	59.1	53.9	46.3	43.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	48.7	51.5	57.2	65.4	70.1	72.4	72.2	67.2	62.7	52.2	49.7	(27)
		MCDB	51.5	51.4	56.9	66.6	74.7	80.3	84.6	85.1	75.6	68.4	54.2	51.3	(28)
	2%	WB	46.0	44.0	48.2	54.2	62.8	66.9	69.1	69.0	64.5	59.2	50.4	47.7	(29)
		MCDB	47.7	46.3	52.3	63.7	73.9	76.5	80.0	80.0	71.2	63.3	52.7	49.6	(30)
	5%	WB	43.4	41.2	45.6	52.2	60.4	64.6	67.3	67.2	62.1	56.7	48.3	45.7	(31)
		MCDB	45.5	43.1	49.3	60.0	69.9	74.2	77.9	76.3	67.9	60.4	50.1	47.5	(32)
	10%	WB	41.5	39.0	43.6	49.9	58.0	62.3	65.3	65.5	60.0	54.5	46.6	43.3	(33)
		MCDB	43.7	40.9	47.3	56.8	66.4	71.0	74.8	73.5	65.1	57.8	48.4	45.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	10.6	11.6	15.2	16.9	16.4	17.3	16.9	14.7	12.5	8.4	7.5	(35)
		MCDBR	8.3	11.7	16.2	20.8	22.0	22.6	23.0	22.2	19.9	15.5	10.2	9.1	(36)
	5% WB	MCWBR	7.3	9.3	11.2	11.6	11.9	11.2	11.2	11.3	11.4	10.3	8.4	8.2	(37)
		MCWBR	8.4	11.2	14.9	19.5	19.8	20.7	21.1	19.8	18.0	14.6	9.7	9.1	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.319	0.344	0.381	0.413	0.411	0.408	0.428	0.422	0.385	0.367	0.343	0.313	(40)
		taud	2.327	2.310	2.234	2.211	2.265	2.294	2.251	2.281	2.361	2.381	2.374	2.331	(41)
	Ebn at Noon	Ebn at Noon	208	239	252	259	266	268	259	254	249	227	197	187	(42)
		Edh at Noon	19	26	35	40	40	39	40	37	30	24	19	16	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	195	412	757	1265	1572	1665	1557	1328	970	532	238	141	(44)
	RadStd	RadStd	22	66	85	170	188	131	211	107	108	67	29	14	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page